

Power-Sample size	Operating Curves: a) $\Phi^2 = \frac{n \sum^a (\tau_i^2)}{a\sigma^2}$ [This quantity is called Φ^2] where $\tau_i = \bar{y}_{i.} - \bar{y}_{..}$ is the i^{th} effect. Given : $a, \sigma^2, v_1 = a - 1, v_2 = a(n - 1)$. The probabilities of type II errors β against Φ can plot for each n. b) If $D = \mu_i - \mu_j$, a desired minimum difference of μ_i, μ_j, then the minimum of Φ^2, $\Phi^2 = \frac{nD^2}{2a\sigma^2}$. Plot probabilities of β vs Φ, for each
--------------------------	---

CHAPTER 4 : 1-BLOCKING FACTOR DESIGN

A. 1-Factor and 1-Block Design

Fixed Effect models (or Means models)

Block Treatment	Bk 1	Bk 2	...	Bk b
Tr 1	y_{11}	y_{12}	...	y_{1b}
Tr 2	y_{21}	y_{22}		y_{2b}
Tr a	y_{a1}	y_{a2}	...	y_{ab}

Model:

$$y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij} \quad [n = 1 \text{ replicate}]$$

$i = 1, \dots, a, \text{ (treatments)}; j = 1, \dots, b \text{ (blocks)},$

$$\text{Assume: } \sum^a \tau_i = 0; \quad \sum^b \beta_j = 0$$

$$\text{Equivalent to: } y_{ij} = \mu_{ij} + \epsilon_{ij} \text{ (Means models)}$$

Parameters: μ_{ij} ($a \times b$ values), σ^2 .

Treat Factor B as Blocking factor, interest in treatment means only.

Random effects model (*) [n replicates]

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

$i = 1, \dots, a \quad j = 1, \dots, b \quad k = 1, \dots, n$

$$\alpha_i \sim N(0, \sigma_\alpha^2) \quad , \quad \beta_j \sim N(0, \sigma_\beta^2) \quad , \quad (\alpha\beta)_{ij} \sim N(0, \sigma_{\alpha\beta}^2)$$

Parameters: $\mu, \sigma_{\alpha}^2, \sigma_{\beta}^2, \sigma_{\alpha\beta}^2, \sigma_e^2$

Effects and error equation/ Sum squares

$$\begin{aligned} \sum_i^a \sum_j^b (y_{ij} - y_{..})^2 &= \sum_i^a \sum_j^b (y_{i.} - y_{..})^2 + \sum_i^a \sum_j^b (y_{.j} - y_{..})^2 + \\ &\quad \text{SS Treatment} + \text{SS Blocking} \\ &\quad + \sum_i^a \sum_j^b (y_{ij} - y_{i.} - y_{.j} + y_{..})^2 \\ &\quad \text{SS Error} \\ &= b(\sum_i^a (y_{i.} - y_{..})^2) + a(\sum_j^b (y_{.j} - y_{..})^2) + \\ &\quad \sum_i^a \sum_j^b (y_{ij} - y_{i.} - y_{.j} + y_{..})^2 \end{aligned}$$

$$SST = SS_{Treatments} + SS_{Block} + SS_{Error}$$

ANOVA

Sources	SS	df	MS	F
Treatments	SSTr	a-1	SSTr/(a-1)	MSTr/MSE
Blocks	SSBk	b-1	SSBk/(b-1)	
Error	SSE	(a-1)(b-1)	SSE/(a-1)(b-1)	
TOTAL	SST	ab -1		

**Expectation of the MS's
Fixed Effects
models**

$$\begin{aligned} E(MSTr) &= \sigma_e^2 + \frac{b(\sum_i^a \tau_i^2)}{a-1} \\ E(MSBk) &= \sigma_e^2 + \frac{a(\sum_j^b \beta_j^2)}{b-1}; \\ E(MSE) &= \sigma_e^2 \end{aligned}$$

Power- Number of blocks b

a) $\Phi^2 = \frac{b(\sum_i^a \tau_i^2)}{a\sigma^2}$, b to be determined.

$(\sum_i^a \tau_i^2) = \text{sum of treatment effects squared.}$

b) D: the desired smallest difference is given, $\Phi^2 = \frac{bD^2}{2a\sigma^2}$, b to be determined.

Given : $D, a, \sigma^2, v_1 = a - 1, v_2 = (a - 1)(b - 1)$

Function Fpower(), package daewr: number of blocks b (need $v_1 = a - 1, v_2 = (a - 1)(b - 1)$)

$$\text{Fpower}(\alpha, \nu_1, \nu_2, nc) \text{ where } nc = \frac{b(\Sigma \tau_i^2)}{\sigma^2}$$

B. Latin Design: 1-Factor and 2-Blocking Factors

Model: $y_{ij} = \mu + \alpha_i + \tau_j + \beta_k + \epsilon_{ijk}$

($n=1$ replicate)

α_i (i^{th} row effect), $i = 1, \dots, p$

τ_j (j^{th} treatment effect), $j = 1, \dots, p$

β_k (k^{th} column effect), $k = 1, \dots, p$

Batches of Raw Material	Operators				
	1	2	3	4	5
1	$A = 24$	$B = 20$	$C = 19$	$D = 24$	$E = 24$
2	$B = 17$	$C = 24$	$D = 30$	$E = 27$	$A = 36$
3	$C = 18$	$D = 38$	$E = 26$	$A = 27$	$B = 21$
4	$D = 26$	$E = 31$	$A = 26$	$B = 23$	$C = 22$
5	$E = 22$	$A = 30$	$B = 20$	$C = 29$	$D = 31$

ANOVA

$$SST = SS_{Rows} + SS_{Treats} + SS_{Columns} + SSE$$

Degree of freedom

$$p^2 - 1 = (p - 1) + (p - 1) + (p - 1) + (p - 2)(p - 1)$$

$$SS_{Rows} = \frac{1}{p} \sum^p (y_{i..} - y_{...})^2$$

$$SS_{Treat} = \frac{1}{p} \sum^p (y_{.j.} - y_{...})^2$$

$$SS_{Column} = \frac{1}{p} \sum^p (y_{..k} - y_{...})^2$$

$$F_0 = \frac{MS_{Treat}}{MSE} \sim F_{p-1, (p-2)(p-1)}$$

C. Balanced Incomplete Blocking Design (BIBD)

a treatments; r replicates

b blocks; each block has k treatments (not all the treatments listed)

Total: $N = ar = bk$

Number of times each pair of treatments appears in same block: $\lambda = \frac{r(k-1)}{a-1}$ or
 $\lambda(a-1) = r(k-1)$

If $a = b$, then the design is symmetric

CHAPTER 5: 2 FACTORS- INTERACTIONS- MULTIFACTOR DESIGNS

(n = number of replicates)

Fixed Effects Models: 2 Factors A and B and possible interaction

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}, \quad k = 1, \dots, n$$

α_i , $i = 1, \dots, a$ (factor A effects)

β_j , $j = 1, \dots, b$ (factor B effects)

$(\alpha\beta)_{ij}$, interaction effect [taking the interaction into account]

Assume: $\sum^a \alpha_i = 0$; $\sum^b \beta_j = 0$; $\sum^{ab} (\alpha\beta)_{ij} = 0$

Or Means model: $y_{ijk} = \mu_{ij} + \epsilon_{ijk}$, $k = 1, \dots, n$ (replicates)

Parameters: μ_{ij} ($a \times b$ values), σ^2

$$\begin{aligned} \sum_i^a \sum_j^b \sum_k^n (y_{ijk} - y_{...})^2 &= \sum_i^a \sum_j^b \sum_k^n (y_{i..} - y_{...})^2 + \sum_i^a \sum_j^b \sum_k^n (y_{.j.} - y_{...})^2 + \\ &\quad \text{Factor A SS} + \text{Factor B SS} + \\ &\quad \sum_i^a \sum_j^b \sum_k^n (y_{ij.} - y_{i..} - y_{.j.} + y_{...})^2 + \sum_i^a \sum_j^b \sum_k^n (y_{ijk} - y_{ij.})^2 \\ &\quad \text{Interaction AB SS} + \text{Error SS} \\ &= bn \left(\sum_i^a (y_{i..} - y_{...})^2 \right) + an \left(\sum_j^b (y_{.j.} - y_{...})^2 \right) + \\ &\quad n \sum_i^a \sum_j^b (y_{ij.} - y_{i..} - y_{.j.} + y_{...})^2 + \sum_i^a \sum_j^b \sum_k^n (y_{ijk} - y_{ij.})^2 \end{aligned}$$

$$SST = SS(A) + SS(B) + SS(AB) + SSE$$

ANOVA

Sources	SS	df	MS	F0
A	SS(A)	a-1	SSA/(a-1)	MSA/MSE
B	SS(B)	b-1	SSB/(b-1)	MSB/MSE
AB	SS(AB)	(a-1)(b-1)	SSAB/(a-1)(b-1)	MAB/MSE
Error	SSE	nab - ab = ab(n-1)	SSE/ ab(n-1)	

Total	SST	nab -1		
-------	-----	--------	--	--

Note: n=1 : Error df=0; MSE , F-stat can't be calculated;

Random effects model (*) $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$ $i = 1, \dots, a \quad j = 1, \dots, b \quad k = 1, \dots, n$ $\alpha_i \sim N(0, \sigma_\alpha^2) \quad , \quad \beta_j \sim N(0, \sigma_\beta^2) \quad , \quad (\alpha\beta)_{ij} \sim N(0, \sigma_{\alpha\beta}^2)$ Parameters: $\mu, \sigma_\alpha^2, \sigma_\beta^2, \sigma_{\alpha\beta}^2, \sigma_e^2$	
Fixed Effects Model Expected MS	$E(MSA) = \sigma_e^2 + \frac{bn(\sum \alpha_i^2)}{a-1}$ $E(MSB) = \sigma_e^2 + \frac{an(\sum \beta_j^2)}{b-1}$ $E(MSAB) = \sigma_e^2 + \frac{n(\sum \alpha \beta (\alpha\beta)_{ij}^2)}{(a-1)(b-1)}$ $E(MSE) = \sigma_e^2$ Under Ho, all MSA, MSB, MSAB are estimates of σ_e^2 . Under Ha, MSA > MSE, MSB > MSE, MSAB > MSE
Random Effects Model Variance components-	In (*) if r.v are independent then $\sigma_y^2 = \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_{\alpha\beta}^2 + \sigma_e^2$ $E(MSA) = \sigma_e^2 + n\sigma_\alpha^2 + (bn)\sigma_{\alpha\beta}^2$ $E(MSB) = \sigma_e^2 + n\sigma_\beta^2 + (an)\sigma_{\alpha\beta}^2$ $E(MSAB) = \sigma_e^2 + n\sigma_{\alpha\beta}^2$ $E(MSE) = \sigma_e^2$ From these, can estimate $\sigma_e^2, \sigma_{\alpha\beta}^2, \sigma_\alpha^2, \sigma_\beta^2$ from the MS
Random Effects Model Estimates For $\sigma_\alpha^2, \sigma_\beta^2, \sigma_{\alpha\beta}^2, \sigma_e^2$	$\sigma_e^2 = MSE$ $\sigma_{\alpha\beta}^2 = \frac{MSAB - MSE}{n}$ $\sigma_\beta^2 = \frac{MSB - MSAB}{an}$ $\sigma_\alpha^2 = \frac{MSA - MSAB}{bn}$